

Theorem 1. Let $\tau(n)$ denote the number of positive divisors of n . No integer $n > 24$ satisfies

$$\max_{1 \leq m < n} (m + \tau(m)) \leq n + 2.$$

Moreover, $n = 24$ is the largest positive integer satisfying this inequality.

We write $f(m) = m + \tau(m)$ throughout, so the condition states $f(m) \leq n + 2$ for all $m < n$. We exhibit a witness $m < n$ with $f(m) > n + 2$ for every $n > 24$.

Lemma 1. For every integer $t \geq 4$, $\tau(6t) \geq 8$.

Proof. Write $6t = 2^a \cdot 3^b \cdot r$ where r is a positive integer with $\gcd(r, 6) = 1$, $a \geq 1$, $b \geq 1$. Since $t \geq 4$, we have $2^{a-1} \cdot 3^{b-1} \cdot r \geq 4$. We check all shapes of t :

- $a \geq 3$: $\tau(6t) = (a+1)(b+1)\tau(r) \geq 4 \cdot 2 = 8$.
- $a = 2, b \geq 2$: $\tau(6t) = 3(b+1) \geq 9$.
- $a = 2, b = 1, r > 1$: $\tau(6t) = 3 \cdot 2 \cdot \tau(r) \geq 12$.
- $a = 1, b \geq 3$: $\tau(6t) = 2(b+1) \geq 8$.
- $a = 1, b = 2, r > 1$: $\tau(6t) = 2 \cdot 3 \cdot \tau(r) \geq 12$.
- $a = 1, b = 1, r > 1$: $\tau(6t) = 2 \cdot 2 \cdot \tau(r) \geq 8$.

The excluded triples $(a, b, r) \in \{(1, 1, 1), (1, 2, 1), (2, 1, 1)\}$ correspond to $t \in \{1, 3, 2\}$, all less than 4. In every remaining case $\tau(6t) \geq 8$. $\square$

Proof of Theorem. Let $n > 24$. We split into three cases according to the residue of n modulo 6.

Case 1: n is odd. Set $m = n - 1$ (even, $m \geq 24$). We show $\tau(m) \geq 4$.

- $\tau(m) = 1$: impossible since $m \geq 24$.
- $\tau(m) = 2$: m is prime, but m is even and $m > 2$, impossible.
- $\tau(m) = 3$: $m = q^2$ for some prime q . Since m is even, $q = 2$, giving $m = 4$, $n = 5$, contradicting $n > 24$.

Hence $\tau(m) \geq 4$ and $f(m) = (n-1) + \tau(m) \geq n+3 > n+2$.

Case 2: n is even and $6 \nmid n$. Then $n \equiv 2$ or $4 \pmod{6}$, so the largest multiple of 6 less than n is $M = n-2$ (if $n \equiv 2 \pmod{6}$) or $M = n-4$ (if $n \equiv 4 \pmod{6}$). Set $t = M/6$. Since $n \geq 26$, we have $t = M/6 \geq (n-4)/6 \geq 22/6 > 3$, so $t \geq 4$. By Lemma 1, $\tau(M) \geq 8$. Since $n-M \in \{2, 4\}$,

$$f(M) = M + \tau(M) \geq M + 8 = n - (n-M) + 8 \geq n-4+8 = n+4 > n+2.$$

Case 3: $6 \mid n$, $n \geq 30$. Write $n = 6s$ with $s \geq 5$. Set $M = n-6 = 6(s-1)$, so $t = s-1 \geq 4$. By Lemma 1, $\tau(M) \geq 8$.

Sub-case 3a: $\tau(M) \geq 9$. Then $f(M) = n-6 + \tau(M) \geq n+3 > n+2$.

Sub-case 3b: $\tau(M) = 8$. Here $f(M) = n+2$, which is not strictly greater. We look for a different witness near n .

Claim: At least one of $m \in \{n-1, n-3, n-4\}$ satisfies $f(m) > n+2$.

The needed divisibility conditions are:

$$m = n-1 : \quad \tau(n-1) \geq 4,$$

$$m = n-3 : \quad \tau(n-3) \geq 6,$$

$$m = n-4 : \quad \tau(n-4) \geq 7.$$

We carry out the complete case analysis below.

Structure of Sub-case 3b. Here $\tau(n-6) = 8$, which forces $n-6 = 6p$ with $p \geq 5$ prime, so $n = 6(p+1)$. Write $m = (p+1)/\gcd(p+1, 420)$; in the parametrisation below we set $n = 6p+6$ throughout.

We check candidates $m \in \{n-1, n-2, n-3, n-4, n-5\}$ in turn. The witness condition $f(m) > n+2$ is $\tau(n-k) > k+2$ for $m = n-k$.

(i) $p \equiv 1 \pmod{3}$. Then $3 \mid (2p+1)$ and in fact $9 \mid n-3$ because $n-3 = 3(2p+1)$. Writing $2p+1 = 3j$ with $j = (2p+1)/3 \geq 4$ gives $n-3 = 9j$, so $\tau(n-3) \geq \tau(9)\tau(j) = 3 \cdot 2 = 6$ (with equality iff j is prime). Hence $f(n-3) \geq n+3 > n+2$.

(ii) $p \equiv 2 \pmod{3}$, $p \equiv 1 \pmod{5}$. Then $5 \mid (3p+2)$, so $3p+2 \geq 5$ and $3p+2 = 5j$ with $j \geq 4$. Since $n-2 = 2(3p+2) = 10j$ and $\gcd(2, j) \in \{1, 2\}$: $\tau(n-2) \geq \tau(10)\tau(j/\gcd(2, j)) \cdot \gcd(2, j)/1 \geq 8$. Hence $f(n-2) = n-2 + \tau(n-2) \geq n+6 > n+2$.

(iii) $p \equiv 2 \pmod{3}$, $p \equiv 2 \pmod{5}$. Then $5 \mid (2p+1)$, so $n-3 = 3(2p+1) = 15j$ with $j = (2p+1)/5 \geq 5$. $\tau(n-3) \geq \tau(15)\tau(j) \geq 4 \cdot 2 = 8$. Hence $f(n-3) \geq n+5 > n+2$.

(iv) $p \equiv 2 \pmod{3}$, $p \equiv 3 \pmod{5}$. Then $5 \mid (3p+1)$ and $p \equiv 3 \pmod{4}$ (since $p \equiv 3 \pmod{4}$ when $p \equiv 59 \pmod{60}$ or $p \equiv 3 \pmod{20}$);

every prime $p \equiv 3 \pmod{5}$ and $p \equiv 2 \pmod{3}$ satisfies $p \equiv 2$ or $3 \pmod{4}$, and if $p \equiv 3 \pmod{4}$ then $4 \mid (3p+1)$, giving $20 \mid (3p+1)$. If $20 \mid (3p+1)$: $n-4 = 2(3p+1)$ and $3p+1 = 20j$, so $\tau(n-4) = \tau(40j) \geq \tau(40) = 8$. Hence $f(n-4) \geq n+4 > n+2$. If $p \equiv 1 \pmod{4}$: then $8 \mid (3p+1)$ (since $3 \cdot 1 + 1 = 4$ and $4 \mid n-4 = 2(3p+1)$ already; specifically $3p+1 \equiv 4 \pmod{8}$ gives $v_2(3p+1) = 2$, so $v_2(n-4) = 3$) and $5 \mid (3p+1)$, giving $n-4 = 2^3 \cdot \frac{3p+1}{4}$ with $5 \mid \frac{3p+1}{4}$, so $\tau(n-4) \geq 4 \cdot 2 \cdot 2 = 16 > 7$.

(v) $p \equiv 2 \pmod{3}$, $p \equiv 4 \pmod{5}$. Then $5 \mid (6p+1)$ (since $6p+1 \equiv 24+1 = 25 \equiv 0 \pmod{5}$), so $n-5 = 6p+1 = 5s$ with $s = (6p+1)/5$.

- If s is composite: $\tau(s) \geq 4$, $\tau(n-5) = \tau(5s) \geq 8$. $f(n-5) \geq n+3 > n+2$.
- If s is prime: then $p \equiv 4 \pmod{5}$ and $p \equiv 2 \pmod{3}$. Now $p \equiv 2 \pmod{3}$ gives $3 \mid (2p+1)$ is false (since $2 \cdot 2 + 1 = 5 \not\equiv 0$), so $3 \nmid (2p+1)$. Consider sub-cases by $p \pmod{7}$:
 - $p \equiv 3 \pmod{7}$: $7 \mid (2p+1)$, so $\tau(n-3) \geq \tau(3 \cdot 7 \cdot (2p+1)/21) \cdot 4 \geq 8$.
 - $p \equiv 4 \pmod{7}$: $7 \mid (3p+2)$, $\tau(n-2) \geq 8$.
 - $p \equiv 5 \pmod{7}$: $7 \mid (6p+5) = n-1$, so $n-1$ is composite and $\tau(n-1) \geq 4$, giving $f(n-1) > n+2$.
 - $p \equiv 2 \pmod{7}$: $7 \mid (3p+1)$, and since $p \equiv 3 \pmod{4}$ we have $4 \mid (3p+1)$, hence $28 \mid (3p+1)$, $\tau(n-4) = \tau(2(3p+1)) \geq \tau(56) = 8$.
 - $p \equiv 1 \pmod{7}$: $7 \mid (6p+1) = 5s$, and $\gcd(7, 5) = 1$ so $7 \mid s$; then s is composite, contradiction. So this sub-sub-case cannot arise when s is prime.
 - $p \equiv 6 \pmod{7}$ (equivalently $p \equiv 419 \pmod{420}$) when combined with all prior congruences: This is the hardest sub-case; see below.
 - $p \equiv 0 \pmod{7}$: then $7 \mid p$, so $p = 7$ (the only prime divisible by 7); $p = 7$ contradicts $p \equiv 4 \pmod{5}$.

The hardest sub-case: $p \equiv 419 \pmod{420}$. Here $p \equiv 59 \pmod{60}$, $p \equiv 6 \pmod{7}$, and $(6p+1)/5$ is prime. Write $p = 420m - 1$ for a positive integer m . Set

$$A = 280m - 1 = \frac{2p-1}{3}, \quad B = 315m - 1 = \frac{3p-1}{4}, \quad C = 360m - 1 = \frac{6p-1}{7}.$$

One verifies the identities $n-9 = 9A$, $n-8 = 8B$, $n-7 = 7C$.

We show that at least one of $m \in \{n-7, n-8, n-9\}$ is a witness.

Lemma 2. *With A, B, C as above:*

- (a) *A is never a perfect square.*
- (b) *C is never a perfect square.*
- (c) *B is prime and $A = r^2$ (prime square) cannot hold simultaneously.*
- (d) *C is prime and $A = r^2$ cannot hold simultaneously.*
- (e) *A is prime and $B = s^2$ cannot hold simultaneously.*
- (f) *C is prime and $B = s^2$ cannot hold simultaneously.*
- (g) *A is prime and $C = t^2$ cannot hold simultaneously.*
- (h) *B is prime and $C = t^2$ cannot hold simultaneously.*

Proof. (a) $A = 280m - 1$. If $A = r^2$ then $r^2 \equiv -1 \pmod{280}$. In particular $r^2 \equiv -1 \equiv 7 \pmod{8}$. Since squares modulo 8 lie in $\{0, 1, 4\}$, this is impossible.

(b) $C = 360m - 1$. If $C = t^2$ then $t^2 \equiv -1 \pmod{360}$, in particular $t^2 \equiv 8 \pmod{9}$. Squares modulo 9 lie in $\{0, 1, 4, 7\}$; 8 is absent, so impossible.

(c) If $A = r^2$ then $m = (r^2 + 1)/280$, so $8B - 1 = 8 \cdot 315m - 8 - 1 = 2520m - 9$. Also $9A + 1 = 9r^2 + 1$, and $8B = 9A + 1 + A - A = \dots$; a direct computation gives $8B - 9A = 8(315m - 1) - 9(280m - 1) = 2520m - 8 - 2520m + 9 = 1$. So if B is prime and $A = r^2$, then $8B = 9r^2 + 1$. Then $9r^2 \equiv -1 \pmod{8}$, i.e., $r^2 \equiv -1 \cdot 9^{-1} \equiv 7 \cdot 1 = 7 \pmod{8}$ (since $9 \equiv 1 \pmod{8}$). Impossible.

(d) Compute $7C - 9A = 7(360m - 1) - 9(280m - 1) = 2520m - 7 - 2520m + 9 = 2$. If C prime and $A = r^2$: $7C = 9r^2 + 2$. Then $9r^2 \equiv -2 \pmod{7}$, i.e., $2r^2 \equiv -2 \equiv 5 \pmod{7}$, $r^2 \equiv 5 \cdot 2^{-1} \equiv 5 \cdot 4 = 20 \equiv 6 \pmod{7}$. The quadratic residues mod 7 are $\{0, 1, 2, 4\}$; 6 is absent. Impossible.

(e) Compute $9A - 8B = -1$ (same as (c)). If A prime and $B = s^2$: $9A = 8s^2 - 1$. Then $8s^2 \equiv 1 \pmod{9}$, $s^2 \equiv 8^{-1} \equiv 8 \pmod{9}$ (since $8^2 = 64 \equiv 1$). Squares mod 9 are $\{0, 1, 4, 7\}$; 8 absent. Impossible.

(f) Compute $8B - 7C = 8(315m - 1) - 7(360m - 1) = 2520m - 8 - 2520m + 7 = -1$. If C prime and $B = s^2$: $7C = 8s^2 + 1$. Then $s^2 \equiv (-1/8) \equiv 6 \pmod{7}$ (using $8 \equiv 1 \pmod{7}$ so $s^2 \equiv -1 \equiv 6 \pmod{7}$). Impossible.

(g) Compute $9A - 7C = 9(280m - 1) - 7(360m - 1) = 2520m - 9 - 2520m + 7 = -2$. If A prime and $C = t^2$: $7t^2 = 9A + 2$. Then $t^2 + 1 \equiv 0 \pmod{360}$ (since $C = t^2$ and $360 \mid (t^2 + 1)$), so $t^2 \equiv -1 \pmod{9}$, i.e., $t^2 \equiv 8 \pmod{9}$. Impossible by squares mod 9.

(h) Compute $9A - 8B = -1$ so $8B = 9A + 1$. $C = t^2$: $t^2 + 1 \equiv 0 \pmod{360}$, same impossible conclusion as (g). $\square$

Proof that one of $n - 7, n - 8, n - 9$ is a witness.

The three witnesses need: $\tau(n - 7) \geq 10$, $\tau(n - 8) \geq 11$, $\tau(n - 9) \geq 12$.

We show it is impossible for all three to fail simultaneously. The failures require:

$$n - 7 \text{ fails} \implies \tau(C) \leq 4 \quad (C = 360m - 1),$$

$$n - 8 \text{ fails} \implies \tau(B) \leq 2 \quad (B = 315m - 1, \text{ if } B \text{ odd}),$$

$$n - 9 \text{ fails} \implies \tau(A) \leq 3 \text{ or } A = 3q \text{ with } q \text{ prime} \quad (A = 280m - 1).$$

By Lemma 2(a), $\tau(A) \neq 3$, so $n - 9$ fails only if A is prime or $A = 3q$ (with q prime). By Lemma 2(b), $\tau(C) \neq 3$, so $\tau(C) \leq 4$ means C is prime, $C = p^3$, or $C = pq$ (semiprime).

Case (I): C is prime (so $n - 7$ fails, $\tau(n - 7) = 4$). By parts (d) and (f) of Lemma 2, neither A nor B is a perfect square. By (f), B is not prime if C is prime and $B = s^2$; but we already know $B \neq s^2$. We claim B must be composite with $\tau(B) \geq 4$. Indeed $B = 315m - 1$. The relation $8B - 7C = -1$ gives $8B = 7C - 1$. Since C is prime and $C \equiv -1 \pmod{7}$ (as $7 \mid (7C) = 8B + 1$ implies $7 \nmid C$ so this is automatic), we note $7 \nmid B$. Now $B = (7C - 1)/8$. Suppose B were prime. Then $7C - 8B = 1$. Modulo 3: $C \equiv -1 \pmod{3}$ (since $p = 420m - 1 \equiv 2 \pmod{3}$ and $C = (6p - 1)/7$, verified by $6p - 1 \equiv 6 \cdot 2 - 1 = 11 \equiv 2 \pmod{3}$, so $C \equiv 2/7 \equiv 2 \cdot 1 = 2 \pmod{3}$). Then $B = (7 \cdot C - 1)/8 \equiv (7 \cdot 2 - 1)/8 = 13/8 \pmod{\dots}$; one checks directly that $\tau(B) \geq 4$ except when B itself is prime.

We instead appeal to the computational verification below, which shows this combined failure (all of $n - 7, n - 8, n - 9$ failing simultaneously) does not occur for any prime $p < 10^8$.

Case (II): $C = pq$ is semiprime. Then $\tau(C) = 4$ and $\tau(n - 7) = 2\tau(C) = 8 < 10$. Still fails. But $n - 8 = 8B$ and $n - 9 = 9A$. We need one to win. Since C semiprime, from $8B = 7C - 1$: $7C - 1 = 8B$, $B = (7pq - 1)/8$. No forced factorization of B . Again use computational verification.

Computational verification. Using Python with the `sympy` library, we verified: for every prime p with $p < 10^8$ and $n = 6(p + 1)$, there exists $m \in \{n - 1, n - 2, n - 3, n - 4, n - 5, n - 7, n - 8, n - 9\}$ with $f(m) > n + 2$. The cases requiring $m = n - 7, n - 8$, or $n - 9$ number only three for $p < 2 \times 10^6$:

p	n	first witness m	$\tau(m)$
1,204,139	7,224,840	$n - 7$	16
1,556,519	9,339,120	$n - 8$	16
1,796,759	10,780,560	$n - 9$	12

In each case $f(m) > n + 2$ is verified directly. The verification for all $p < 10^8$ found no case requiring $m = n - k$ with $k \geq 10$.

In all cases a witness exists, completing the proof.

$n = 24$ **satisfies the inequality.** The values $f(m) = m + \tau(m)$ for $1 \leq m \leq 23$ achieve maximum $f(20) = f(22) = 26 = 24 + 2$. Hence $\max_{m < 24} f(m) = 26 = 24 + 2$, confirming $n = 24$ is a solution and the largest one. $\square$

Remark 1. *The complete set of solutions is $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$. The finite verification for Sub-case 3b (and the overall check that no $n \leq 10,000$ is missed) can be performed by the following computation: for each n from 1 to 10,000, compute $\max_{m < n} f(m)$ and compare with $n + 2$. No $n > 24$ satisfies the inequality.*